Question

Show that the normal distribution is the limiting form of the poisson distribution.

Sol The Normal distribution is the limiting form of the poisson distribution when $\mu \to \infty$.

The p.m.f of the poisson distribution is.

$$p(X=x) = \frac{e^{-\mu} \cdot (\mu)^x}{x!} \quad x = 0, 1, 2, \ldots, \infty$$

$$E(x) = V(x) = \mu$$

let $$Z = \frac{x - E(x)}{\sqrt{V(x)}} \Rightarrow \frac{x-\mu}{\sqrt{\mu}}$$

$$E(Z) = 0, \quad V(Z) = 1$$

In order to show that $\mu \to \infty$, the limiting distribution of Z is standard normal distribution.

we have m.g.f by definition

$$M_Z(t) = E(e^{tz}) \Rightarrow E\left(e^{t\left(\frac{x-\mu}{\sqrt{\mu}}\right)}\right)$$

$$= e^{-t\sqrt{\mu}} E\left(e^{\frac{tx}{\sqrt{\mu}}}\right)$$

$M_x(t) = E(e^{tx})$

$= \sum e^{tx} p(x)$

$$= e^{-t\sqrt{\mu}} \cdot \sum_{x=0}^{\infty} e^{\frac{tx}{\sqrt{\mu}}} p(x)$$

$$= e^{-t\sqrt{\mu}} \cdot \sum_{x=0}^{\infty} e^{\frac{tx}{\sqrt{\mu}}} \cdot \frac{e^{-\mu} \cdot \mu^x}{x!}$$

$$= e^{-t\sqrt{\mu}} \cdot e^{-\mu} \sum_{x=0}^{\infty} \frac{\left(\mu \cdot e^{\frac{t}{\sqrt{\mu}}}\right)^x}{x!}$$

$$= e^{-t\sqrt{\mu}} \cdot e^{-\mu} \sum_{x=0}^{\infty} \frac{\left(\mu e^{t/\sqrt{\mu}}\right)^x}{x!}$$

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$$M_Z(t) = e^{-t\sqrt{\mu}} \cdot e^{-\mu} e^{\mu e^{t/\sqrt{\mu}}}$$

$$= e^{-t\sqrt{\mu}} \cdot e^{-\mu + \mu e^{t/\sqrt{\mu}}}$$

$$= e^{-t\sqrt{\mu}} \cdot e^{\mu(e^{t/\sqrt{\mu}} - 1)}$$

taking log both sides.

$$\ln M_Z(t) = -t\sqrt{\mu} + \mu(e^{t/\sqrt{\mu}} - 1)$$

$\frac{3}{2} - 1$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\ln M_Z(t) = -t\sqrt{\mu} + \mu\left[\cancel{1} + \frac{t}{\sqrt{\mu}} + \frac{1}{2!}(t/\sqrt{\mu})^2 + \frac{1}{3!}\frac{t^3}{\mu^{\frac{3}{2}}} \cancel{-1}\right]$$

$$= \cancel{-t\sqrt{\mu}} + \cancel{t\sqrt{\mu}} + \frac{1}{2!}t^2 + \frac{1}{3!}\frac{t^3}{\sqrt{\mu}} + \cdots$$

$\mu \to \infty$

$$\lim_{\mu\to\infty} \ln M_Z(t) = \frac{1}{2!}t^2 \Rightarrow \frac{1}{2}t^2$$

$$\lim_{\mu\to\infty} M_Z(t) = e^{\frac{1}{2}t^2}$$

which is the m.g.f. of the standard normal distribution.

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Question
State and prove the central limit theorem

Statement: Let $f(x)$ be a density with mean 'μ' and variance σ^2. Let $\bar{x}$ be the mean of a random sample of size 'n' from $f(x)$. Let the R.V 'z' be defined as

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

The density of 'z' approaches to the normal distribution with mean zero and variance 1. As n increases i.e $n \to \infty$

Proof: we shall prove the theorem under the assumption that the distribution has the m.g.f. we have

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$\because \quad \bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

$$z = \frac{(x_1 + x_2 + \dots + x_n)/n - \mu}{\sigma/\sqrt{n}}$$

Let $x_1 + x_2 + \dots + x_n = y$

$$z = \frac{y/n - \mu}{\sigma/\sqrt{n}} = \frac{\frac{y - n\mu}{n}}{\sigma/\sqrt{n}}$$

$$= \frac{y - n\mu}{\sigma\sqrt{n}}$$

using the property of m.g.f

$\because M_{(x_1+x_2+\cdots+x_n)}^{t} = M_{x_1}(t) \cdot M_{x_2}(t) \cdots M_{x_n}(t)$ (20)

but $M_{x_1}(t) = M_{x_2}(t) = \cdots = M_{x_n}(t) = M_x(t)$

$$M_y(t) = [M_x(t)]^n \text{ --- (1)}$$

Now

$$M_z(t) = E(e^{tz})$$

$$= E\left[e^{t\left(\frac{y-n\mu}{\sigma\sqrt{n}}\right)}\right]$$

$$= e^{-\frac{n\mu t}{\sigma\sqrt{n}}} E\left[e^{ty/\sigma\sqrt{n}}\right]$$

$$= e^{-\frac{\sqrt{n}\mu t}{\sigma}} M_y\left(\frac{t}{\sigma\sqrt{n}}\right)$$

using equation (1)

$$M_z(t) = e^{-\frac{\sqrt{n}\mu t}{\sigma}} \left[M_x\left(\frac{t}{\sigma\sqrt{n}}\right)\right]^n$$

By taking log both sides.

$$\ln M_z(t) = -\frac{\sqrt{n}\mu t}{\sigma} + n \ln M_x\left(\frac{t}{\sigma\sqrt{n}}\right)$$

By expanding $M_x\left(\frac{t}{\sigma\sqrt{n}}\right)$ as a power series in 't' i.e $M_0(t) = 1 + \mu_1' t + \mu_2' \frac{t^2}{2!} + \cdots$

$$\ln M_z(t) = -\frac{\sqrt{n}\mu t}{\sigma} + n \ln\left[1 + \mu_1' \frac{t}{\sigma\sqrt{n}} + \mu_2' \frac{(t/\sigma\sqrt{n})^2}{2!} + \mu_3' \frac{\left(\frac{t}{\sigma\sqrt{n}}\right)^3}{3!} + \cdots\right]$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

$$\ln M_Z(t) = -\frac{\mu\sqrt{n}}{\sigma}t + n\Big[\Big\{\mu_1'\frac{t}{\sigma\sqrt{n}} + \frac{\mu_2'}{2!}\Big(\frac{t}{\sigma\sqrt{n}}\Big)^2 + \frac{\mu_3'}{3!}\Big(\frac{t}{\sigma\sqrt{n}}\Big)^3 + \Big\}$$

$$-\frac{1}{2}\Big\{\mu_1'\Big(\frac{t}{\sigma\sqrt{n}}\Big) + \frac{\mu_2'}{2!}\Big(\frac{t}{\sigma\sqrt{n}}\Big)^2 + \frac{\mu_3'}{3!}\Big(\frac{t}{\sigma\sqrt{n}}\Big)^3 + \Big\}^2$$

$$+\frac{1}{3}\Big\{\mu_1'\Big(\frac{t}{\sigma\sqrt{n}}\Big) + \frac{\mu_2'}{2!}\Big(\frac{t}{\sigma\sqrt{n}}\Big)^2 + \frac{\mu_3'}{3!}\Big(\frac{t}{\sigma\sqrt{n}}\Big)^3 + \Big\}^3$$

$$-\frac{1}{4}\Big\{\mu_1'\Big(\frac{t}{\sigma\sqrt{n}}\Big) + \frac{\mu_2'}{2!}\Big(\frac{t}{\sigma\sqrt{n}}\Big)^2 + \frac{\mu_3'}{3!}\Big(\frac{t}{\sigma\sqrt{n}}\Big)^3 + \dots\Big\}^4\Big]$$

By collecting the powers of 't' we get

$$\ln M_Z(t) = \Big(-\frac{\mu\sqrt{n}}{\sigma} + \mu_1'\frac{\sqrt{n}}{\sigma}\Big)t + \Big(\frac{\mu_2'}{2\sigma^2} - \frac{\mu_1'^2}{2\sigma^2}\Big)t^2$$

$$+\Big(\frac{\mu_3'}{3!\sigma^3\sqrt{n}} - \frac{\mu_1'\mu_2'}{2\sigma^3\sqrt{n}} + \frac{\mu_1'^3}{3\sigma^3\sqrt{n}}\Big)t^3 + \dots$$

Since $\mu = \mu_1'$

$\mu_2' - (\mu_1')^2 = \sigma^2$

$$= 0 + \frac{1}{2\sigma^2}(\mu_2' - \mu_1'^2)t^2 + \Big(\frac{\mu_3'}{6} - \frac{\mu_1'\mu_2'}{2} + \frac{\mu_1'^3}{3}\Big)\frac{t^3}{\sigma^3\sqrt{n}}$$

$$= 0 + \Big(\frac{1}{2\cancel{\sigma^2}}\cdot\cancel{\sigma^2}\Big)t^2 + \frac{t^3}{\sigma^3\sqrt{n}}$$

As $n \to \infty$

$$\lim_{n\to\infty} \ln M_Z(t) = \frac{1}{2}t^2$$

$$\lim_{n\to\infty} M_Z(t) = e^{\frac{1}{2}t^2}$$

This is the moment generating function of the Standard normal Distribution

Chebyshev's Inequality

State and prove the chebyshev's inequality

Statement If 'x' is a r.v having mean 'μ' and Variance 'σ^2', and 'K' is any positive constant, Then the probability that a value of 'x' falls within 'Kσ' of the means is atleast $1-\frac{1}{K^2}$ i.e

$$P(\mu-K\sigma < x < \mu+K\sigma) = 1-\frac{1}{K^2}$$

$$P(|x-\mu| \geq K\sigma) \leq \frac{1}{K^2}$$

Proof By definition

$$\sigma^2 = E(x-\mu)^2 = \int_{-\infty}^{\infty}(x-\mu)^2 f(x)\,dx$$

$$\sigma^2 = \int_{-\infty}^{\mu-K\sigma}(x-\mu)^2 f(x)\,dx + \int_{\mu-K\sigma}^{\mu+K\sigma}(x-\mu)^2 f(x)\,dx + \int_{\mu+K\sigma}^{\infty}(x-\mu)^2 f(x)\,dx$$

$$\geq \int_{-\infty}^{\mu-K\sigma}(x-\mu)^2 f(x)\,dx + \int_{\mu+K\sigma}^{\infty}(x-\mu)^2 f(x)\,dx \quad \text{--- (1)}$$

Now since

$|x-\mu| \geq K\sigma$ whenever

$x \geq \mu+K\sigma$

$x \leq \mu-K\sigma$

By squaring we have

$(x-\mu)^2 \geq K^2\sigma^2$

$$\sigma^2 \geq \int_{-\infty}^{\mu-K\sigma}(K^2\sigma^2) f(x)\,dx + \int_{\mu+K\sigma}^{\infty}(K^2\sigma^2) f(x)\,dx$$

$$\frac{\sigma^2}{k^2\sigma^2} \geq \int_{-\infty}^{\mu-k\sigma} f(x)dx + \int_{\mu+k\sigma}^{\infty} f(x)dx$$

$$\frac{1}{k^2} \geq P(\mu-k\sigma < x < \mu+k\sigma)$$

$$P(\mu-k\sigma < x < \mu+k\sigma) \leq \frac{1}{k^2}$$

$$P(\mu-k\sigma < x < \mu+k\sigma) \geq 1-\frac{1}{k^2}$$

Question

Suppose that, it is known that the no of items produced in a factory during a week is a R.V with mean = 50. If the variance of the week production is equal to 25. Then what can be said about the probability that this week production is between 40 and 60.

Sol

$$P(\mu-k\sigma < x < \mu+k\sigma) = 1-\frac{1}{k^2}$$

$$P(40 < x < 60) = ?$$

Given
$\mu = 50$
$\sigma^2 = 25$
$\sigma = 5$

it means $\mu - k\sigma = 40$ — (i)
$\mu + k\sigma = 60$ — (ii)

$$2\mu = 100$$
$$\mu = 50$$

$$50 - k(5) = 40$$
$$50 - 40 = 5k$$
$$k = 2$$

Now

$p(40 < x < 60) = 1 - \frac{1}{2^2}$

$p(40 < x < 60) = 3/4$

So, the probability that this week production will be between 40 and 60 is at least 0.75

Question

If $X \sim N(0,1)$ find the distribution of X^2

Sol we know that

$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \quad -\infty \le x < \infty$

even function

$dF(x) = \frac{2}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx \quad 0 < x < \infty$

put $x^2 = y \quad x \to 0 \quad y = 0$

$x = \sqrt{y} \quad x \to \pm\infty \quad y = \infty$

$2x\,dx = dy$ therefore

$0 < y < \infty$

$dx = \frac{dy}{2x} = dy/2\sqrt{y}$

$dF = \frac{2}{\sqrt{2\pi}} e^{-y/2} \frac{dy}{2\sqrt{y}} \quad -\infty < y < \infty$

$dF = \frac{\cancel{2}}{\cancel{2}\sqrt{2\pi}} e^{-y/2} y^{-\frac{1}{2}} dy$ Due to square $0 < y < \infty$

put $y/2 = m$

$y = 2m$

$dy = 2dm$

$\sqrt{2}\sqrt{2} = 2$

$dF = \frac{1}{\sqrt{\cancel{2}\pi}} e^{-m} (\cancel{2}m)^{-\frac{1}{2}} \cancel{2} dm \Rightarrow \frac{1}{\sqrt{\pi}} e^{-m} (m)^{-\frac{1}{2}} dm$

$$dF = \frac{1}{\lceil \frac{1}{2}} e^{-\frac{m}{1}} m^{\frac{1}{2}-1} dm$$

which is a gamma distribution with parameter $\frac{1}{2}$

~~and p~~

$$\boxed{f(\alpha,\beta) = \frac{1}{\lceil \alpha \, \beta^{\alpha}} x^{\alpha-1} e^{-\frac{x}{\beta}} \qquad 0 < x < \infty}$$

Note: on the back of the page.

Assignment

Q=1: If x is normally distributed with mean zero and variance unity what is the expectation and variance of (i) x^2 (ii) e^{ax}

(i) $\mu_1' = E(x^2) = 1$, $\mu_2 = V(x^2) = 2$ $\quad f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$ $-\infty < x < \infty$

(ii) $E(e^{ax}) = e^{a^2/2}$ $\quad V(x) = e^{2a^2} - e^{a^2}$

Q=2 obtain the characteristic function of a normal variate.

$$\phi_x(t) = E(e^{itx}) = \int_{-\infty}^{\infty} e^{itx} f(x)\, dx$$

$$\phi_x(t) = e^{i\mu t - \frac{1}{2}\sigma^2 t^2}$$

Q=3 For a normal distribution show that

$$\mu_{2n+2} = \sigma^2 \mu_{2n} + \sigma^3 \frac{d}{d\sigma} \mu_{2n}$$

μ_n is the nth centeral moment about mean. Also find the value of β_2

Question
State and prove the inversion theorem

Statement
If $F(x)$ and $\phi(t)$ denotes respectively the distribution function and characteristic function of the r.v 'x'. If $(a+h)$ and $(a-h)$ where $(h>0)$ are continuity points of the distribution function $F(x)$.
when

$$F(a+h)-F(a-h) = \lim_{T\to\infty} \frac{1}{\pi}\int_{-T}^{T} \frac{\sin ht}{t} e^{-ita}\phi(t)\,dt$$

proof we give the prove only for a r.v of continuous type with density function $f(x)$
let us denote

$$J = \frac{1}{\pi}\int_{-T}^{T} \frac{\sin ht}{t} e^{-ita}\phi(t)\,dt \quad \text{--- (1)}$$

From the definition of characteristic function

$$\phi(t) = E(e^{itx}) = \int_{-\infty}^{\infty} e^{itx} f(x)\,dx$$

$$J = \frac{1}{\pi}\int_{-T}^{T}\left[\int_{-\infty}^{\infty} \frac{\sin ht}{t} e^{-ita}\cdot e^{itx} f(x)\,dx\right]dt$$

$$J = \frac{1}{\pi}\int_{-\infty}^{\infty}\left[\int_{-T}^{T} \frac{\sin ht}{t} e^{it(x-a)} f(x)\,dt\right]dx$$

$$e^{ix} = \cos x + i\sin x$$

$$J = \frac{1}{\pi}\int_{-\infty}^{\infty}\left[\int_{-T}^{T} \frac{\sin ht}{t} \{\underbrace{\cos t(x-a)}_{\text{even function}}\} + \underbrace{i\sin(x-a)\}}_{\text{odd function}} f(x)\,dt\right]dx$$

$$J = \frac{2}{\pi}\int_{-\infty}^{\infty}\left[\int_{0}^{T} \frac{\sin ht}{t}\cos\{t(x-a)\} f(x)\,dt\right]dx$$

$\because$ $\sin A \cos B = \frac{1}{2}\left[\sin(A+B) + \sin(A-B)\right]$

Here $A = ht$

$B = (x-\alpha)t$

$$J = \int_{-\infty}^{\infty}\left[\frac{2}{\pi}\cdot\frac{1}{2}\left\{\int_0^{T}\frac{\sin(x-\alpha+h)t}{t} - \int_0^{T}\frac{\sin(x-\alpha-h)t}{t}\right\}dt\right] f(x)\,dx$$

$$= \int_{-\infty}^{\infty}\left[\frac{1}{\pi}\int_0^{T}\frac{\sin(x-\alpha+h)t}{t}dt - \frac{1}{\pi}\int_0^{T}\frac{\sin(x-\alpha-h)t}{t}dt\right] f(x)\,dx$$

$$J = \int_{-\infty}^{\infty} g(x,T)\, f(x)\, dx \quad \text{—— (2)}$$

$$\lim_{T\to\infty}\frac{1}{\pi}\int_0^{T}\frac{\sin\alpha t}{t}dt = \begin{cases}\frac{1}{2} & \alpha > 0\\ -\frac{1}{2} & \alpha < 0\end{cases}$$

Here $|\alpha| = |x-\alpha+h| > \delta > 0$

$$\lim_{T\to\infty} g(x,T) = \begin{cases} 0 & \text{for } x < \alpha - h\\ -1/2 & x = \alpha - h\\ 1 & \alpha - h < x < \alpha + h\\ +1/2 & x = \alpha + h\\ 0 & x > \alpha + h\end{cases}$$

$F(\frac{1}{2}) - F(-\frac{1}{2})$

$\frac{1}{2} + \frac{1}{2} = 1$

$-1/2$ $\quad$ $1/2$

$\alpha - h$ $\quad$ $\alpha + h$

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from equation (2)

$$\lim_{T\to\infty}\int_{-\infty}^{\infty} g(x;t)\, f(x)\, dx$$

$$= \int_{\alpha-h}^{\alpha+h} f(x)\, dx.$$

$$\lim_{T\to\infty} J = F(\alpha+h) - F(\alpha-h) \quad \text{—— (3)}$$

From eq (1) and (3)

$$F(\alpha+h) - F(\alpha-h) = \lim_{T\to\infty} \frac{1}{\pi}\int_{-T}^{T} \frac{\sin ht}{t} e^{-it\alpha} \phi(t)\, dt$$

Question

State and prove the uniqueness theorem

Statement

characteristic function uniquely determine the distribution i.e a necessary and sufficient condition for two distributions with p.d.f $f_1(\cdot)$ and $f_2(\cdot)$, to be identical is that their characteristic functions are identical i.e

$$\phi_1(t) = \phi_2(t) = \phi(t)$$

Proof. If $f_1(\cdot)$ and $f_2(\cdot)$ are equal.

$$f_1(\cdot) = f_2(\cdot)$$

Then from the definition of characteristic function

$$\phi_1(t) = \int_{-\infty}^{\infty} e^{itx} f_1(x)\, dx = \int_{-\infty}^{\infty} e^{itx} f_2(x)\, dx = \phi_2(t)$$

If the characteristic function is absolutely integrable over the interval $-\infty$ to $+\infty$. then the corresponding density function $f(x)$ can be determined by $\phi(t)$

In fact. The absolute integrability of $\phi(t)$. It follows that the improper integral.

$$F(x+h) - F(x-h) = \lim_{T\to\infty} \frac{1}{\pi}\int_{-T}^{T} \frac{\sin ht}{t} e^{-itx} \phi(t)\, dt$$

Dividing both sides by $2h$, we get

$$\frac{F(x+h)-F(x-h)}{2h} = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{\sin ht}{ht} \cdot e^{-itx} \varphi(t)\, dt$$

where '$x+h$' and '$x-h$' are the continuity points of $F(x)$ where $h \to 0$

Now

$$\lim_{h\to 0} \frac{F(x+h)-F(x-h)}{2h} = \lim_{h\to 0} \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{\sin ht}{ht} e^{-itx} \varphi(t)\, dt$$

integral on the right side is bounded by an integrable function and hence dominate convergance.

$$\lim_{h\to 0} \frac{F(x+h)-F(x-h)}{2h} = \frac{1}{2\pi}\int_{-\infty}^{\infty} \lim_{h\to 0} \frac{\sin ht}{ht} e^{-itx} \varphi(t)\, dt$$

$$\because \lim_{\theta\to 0} \frac{\sin\theta}{\theta} = 1.$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-itx} \varphi(t)\, dt$$

By mean value Theorem of Differential calculas

$$\lim_{h\to 0} \frac{F(x+h)-F(x-h)}{2h} = F'(x) = f(x)$$

So, $$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-itx} \varphi(t)\, dt$$

Question: The characteristic function of a r.v 'x' is given by the formula.

$$\varphi(t) = e^{-t^2/2}$$

Find the density function of r.v 'x'.

Sol As we know that

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-itx}\,\phi(t)\,dt$$

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-itx}\cdot e^{-t^2/2}\,dt$$

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-itx-t^2/2}\,dt.$$

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-\frac{1}{2}(2itx+t^2)}\,dt$$

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-\frac{1}{2}\left[(t^2+2itx+(ix)^2-(ix)^2\right]}\,dt$$

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-\frac{1}{2}\left[(t+ix)^2-(ix)^2\right]}\,dt$$

$$= \frac{e^{\frac{(ix)^2}{2}}}{2\pi}\int_{-\infty}^{\infty} e^{-\frac{1}{2}(t+ix)^2}\,dt$$

$$= \frac{1\,e^{\frac{(ix)^2}{2}}}{\sqrt{2\pi}}\cdot\underbrace{\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-\frac{1}{2}(t+ix)^2}\,dt}_{1}$$

density function of normal Distribution with mean (ix) and Variance 1

$$f(x) = \frac{1}{\sqrt{2\pi}}\,e^{\frac{(ix)^2}{2}}$$

Question

State and prove the law of large numbers

Statement

Let $x_1, x_2, \ldots, x_n$ be independent trials, process with $\mu = E(x_j)$ and $S_n = x_1 + x_2 + \cdots + x_n$. Then for any $t > 0$.

$$P\left[\left|\frac{S_n}{n} - \mu\right| \geq t\right] \to 0 \quad \text{when } n \to \infty$$

$$P\left[\left|\frac{S_n}{n} - \mu\right| < t\right] \to 1 \quad \text{As } n \to \infty.$$

where $\frac{S_n}{n} = \bar{x}$ and t is any small number. It is also called weak law of large numbers.

Proof Since $x_1, x_2, \ldots, x_n$ be independent trials and have same densities so.

$$S_n = \sum x_i$$

$$V(S_n) = \sum_{i=1}^{n} V(x_i) = nV(x_i)$$

$$Var(S_n) = nVar(x_i) = n\sigma^2$$

Mean $$E\left(\frac{S_n}{n}\right) = \frac{1}{n} E(S_n) = \frac{nE(x_i)}{n}$$

$$E\left(\frac{S_n}{n}\right) = E(x_i) = \mu$$

Variance $$V\left(\frac{S_n}{n}\right) = \frac{1}{n^2} V(S_n) \Rightarrow \frac{n\sigma^2}{n^2} \Rightarrow \frac{\sigma^2}{n}$$

By chebyshev's inequality.

$$P\left[\left|\frac{S_n}{n} - \mu\right| \geq t\right] = \frac{Var\left(\frac{S_n}{n}\right)}{t^2}$$

$$P\left[\left|\frac{S_n}{n} - \mu\right| \geq t\right] = \frac{\sigma^2/n}{t^2}$$

$$P\left[\left|\frac{S_n}{n} - \mu\right| \geq t\right] = \frac{\sigma^2}{nt^2}$$

$$P\left[\left|\frac{S_n}{n} - \mu\right| \geq t\right] \to 0 \quad n \to \infty$$

or

$$P\left[\left|\frac{S_n}{n} - \mu\right| < t\right] \to 1 \quad n \to \infty.$$

Question

Let $X_1, X_2, \ldots$ be iid with common law $b(1, p)$ holds the law of large no.

Sol

$$X_i \sim b(1, p)$$

$$E(X_i) = p$$

$$V(X_i) = pq$$

we know that

$$P\left[\left|\frac{S_n}{n} - \mu\right| \geq t\right] = \frac{\sigma^2}{nt^2}$$

$$= \frac{pq}{nt^2}$$

as $n \to \infty$.

$$P\left[\left|\frac{S_n}{n} - \mu\right| \geq t\right] \to 0$$
